Supplementary Information

Supporting Information for:

Differentiating Clinical Pathogenicity from Asymptomatic Colonization in Diarrhoeagenic *Escherichia coli*: A Systematic Review and Hierarchical Meta-Analysis

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- **S1 Fig. Methodological Quality and Risk of Bias Dashboard (NOS & JBI appraisal).** Itemized methodological quality scoring across 68 African surveillance cohorts and 31 global case-control cohorts evaluated via the Newcastle–Ottawa Scale (NOS) and Joanna Briggs Institute (JBI) critical appraisal checklist.
- **S2 Fig. Continental African DEC Pathotype Prevalence Synthesis.** Multi-panel synthesis displaying pooled random-effects prevalence estimates, Freeman–Tukey double-arcsine distributions, and between-study heterogeneity for EAEC, ETEC, EPEC, STEC, EIEC, and DAEC across primary African surveillance cohorts.
- **S3 Fig. Subregional and Temporal Dynamics of African DEC Pathotypes.** Small-multiples comparative analysis evaluating pathotype shifts across five UN African subregions, longitudinal decadal trends (2000–2026), and the epidemiological transition from typical EPEC (*bfpA*+) to atypical EPEC (*bfpA*-).
- **S4 Fig. Diagnostic Platform Detection Gradients and Discordance.** Empirical diagnostic yield escalation across culture, biochemical phenotyping, conventional PCR, qPCR/TAC, and WGS; WHO syndromic recovery boxplots; and primer-binding site polymorphism discordance.
- **S5 Fig. Two-Way Hybrid DEC Architecture and Multi-Pathotype Overlap Network.** Circos bilateral mobilome chord diagram and UpSet multi-pathotype intersection matrix quantifying co-occurrence frequencies across 9,360 African children, demonstrating the restriction of hybrids to two-way configurations.
- **S6 Fig. Matched Case-Control Pathotype Disease Attribution Across African Cohorts.** Paired trajectory slope-lollipop plots and random-effects odds ratios resolving true etiological disease attribution from asymptomatic colonization across 13 matched case-control studies.
- **S7 Fig. Global DEC Virulome Meta-Analysis and Locus-Level Disease Attributions.** Forest plots and carrier frequency distributions across 29 discrete virulence determinants spanning EAEC SPATEs and adhesins (*aggR*, *aatA*, *aaiC*, *aap*, *sepA*, *pet*, *pic*, *astA*), EPEC locus of enterocyte effacement and OI-122 effectors (*eae*, *bfpA*, *lifA*, *nleB*, *nleE*), and mucosal fitness/iron-acquisition systems (*fyuA*, *irp2*, *chuA*).
- **S8 Fig. Cross-Continental Virulence Loci Invariance and Comparative Matrix.** Locus-by-locus disease attribution heatmap comparing effect sizes (log2 Odds Ratio) across five continents, confirming invariant clinical pathogenicity profiles independent of geographical origin.

- **S9 Fig. Host Developmental Age-Ontogeny Switch in DAEC and EPEC Susceptibility.** Age-stratified case-control odds ratios demonstrating infant resistance (<12 months) versus toddler vulnerability (12–59 months) for DAEC (*daaC*), contrasted with peak infant vulnerability to typical EPEC (*bfpA*).
- **S10 Fig. Small-Study Effects, Publication Bias Diagnostics, and Funnel Plots.** Funnel plots, Egger's weighted linear regression, Begg's rank test, and trim-and-fill sensitivity models across all 29 virulence loci and 6 pathotype prevalence tiers.

Index of Supplementary Tables

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- **S2 Table. Electronic Database Search Strategies.** Comprehensive search strings, boolean operators, and retrieval tallies across PubMed/MEDLINE, Scopus, Web of Science, Embase, EBSCOhost, and AJOL.
- **S3 Table. Methodological Quality and Risk of Bias Evaluation (NOS & JBI Scores).** Itemized quality appraisal and risk of bias scores for all included African surveillance and global case-control cohorts.
- **S4 Table. Primary African DEC Cohorts Included in the Surveillance Synthesis (k=68).** Demographic, clinical, diagnostic, and pathotype recovery metadata across all 68 African surveillance cohorts.
- **S5 Table. Subregional and Diagnostic Modality Pooled Prevalence Breakdown.** Stratified random-effects meta-analyses detailing Freeman–Tukey double-arcsine pooled proportions, 95% confidence intervals, and heterogeneity statistics.
- **S6 Table. Univariate and Multivariate Meta-Regression Models of Between-Study Heterogeneity.** REML variance partitioning models evaluating the influence of publication era, diagnostic modality, and host age strata.
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- **S8 Table. Subgroup and Sensitivity Analyses of Virulence Loci.** Leave-one-out sensitivity models, diagnostic platform stratifications, and publication bias diagnostics across all evaluated loci.
- **S9 Table. Comparative Methodological Framework Benchmarking Against Prior Literature.** Five-dimension architectural comparison benchmarking the present study against prior regional enteric syntheses.

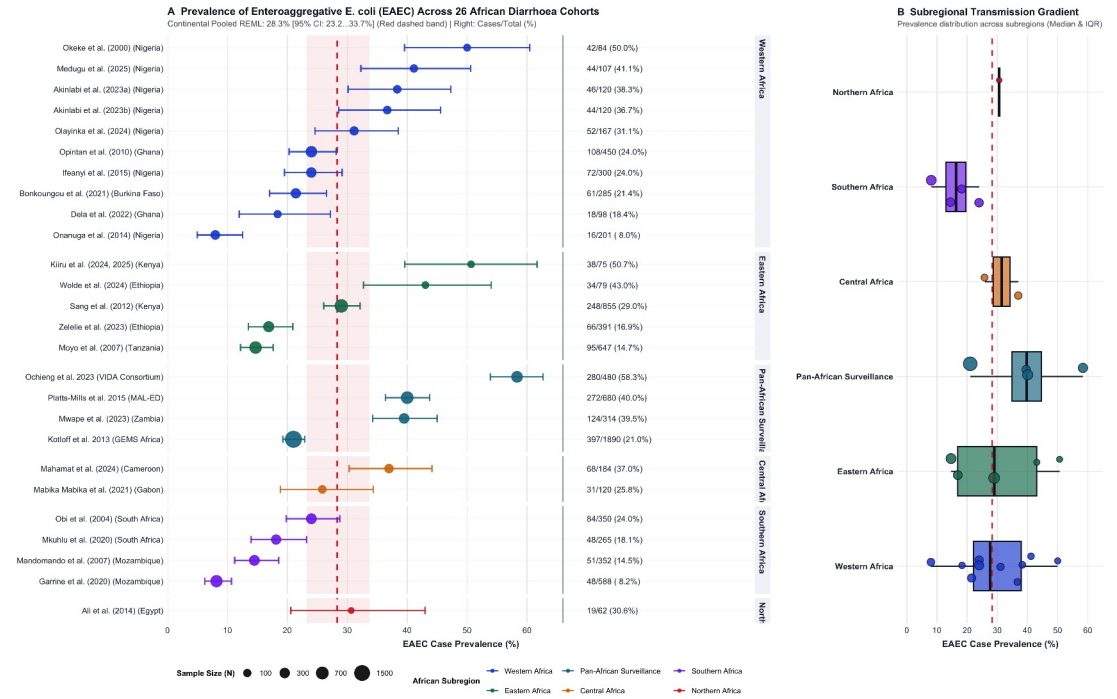
Detailed Supplementary Figures

S1 Fig

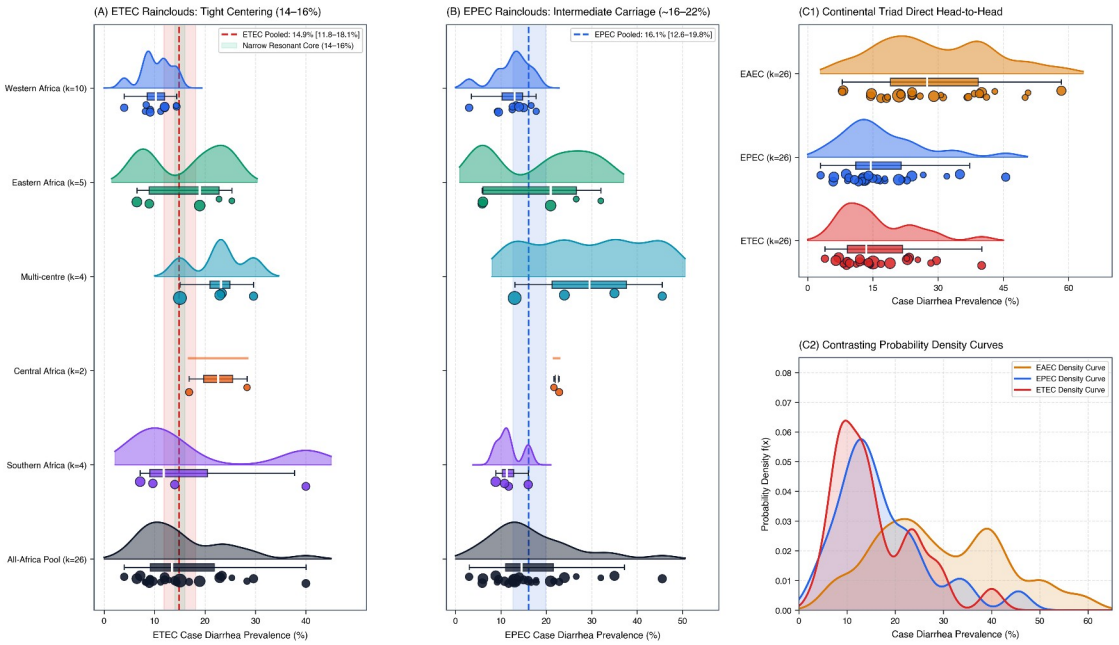


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S2 Fig



Supplementary Figure S2: Comparative Raincloud Distributions of Enterotoxigenic (ETEC) and Enteropathogenic (EPEC) E. coli Demonstrating ETEC's Continental Centering (14–16%) vs EAEC's Bimodal Dispersion (28–35%) Across African Surveillance Sites



Supplementary Figure S15: Multivariate Cohort Ordination Biplot: 6-Pathotype Diagnostic Landscape Across 28 African Cohorts Demonstrating Superior Cryptic Pathotype & Hybrid Resolution by WGS and qPCR over Conventional Gel-Based PCR Surveillance

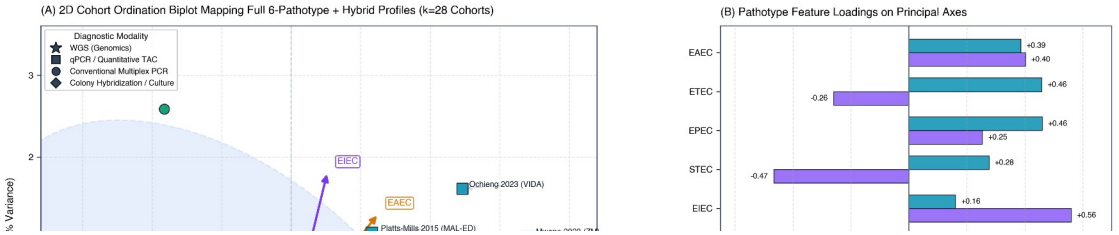
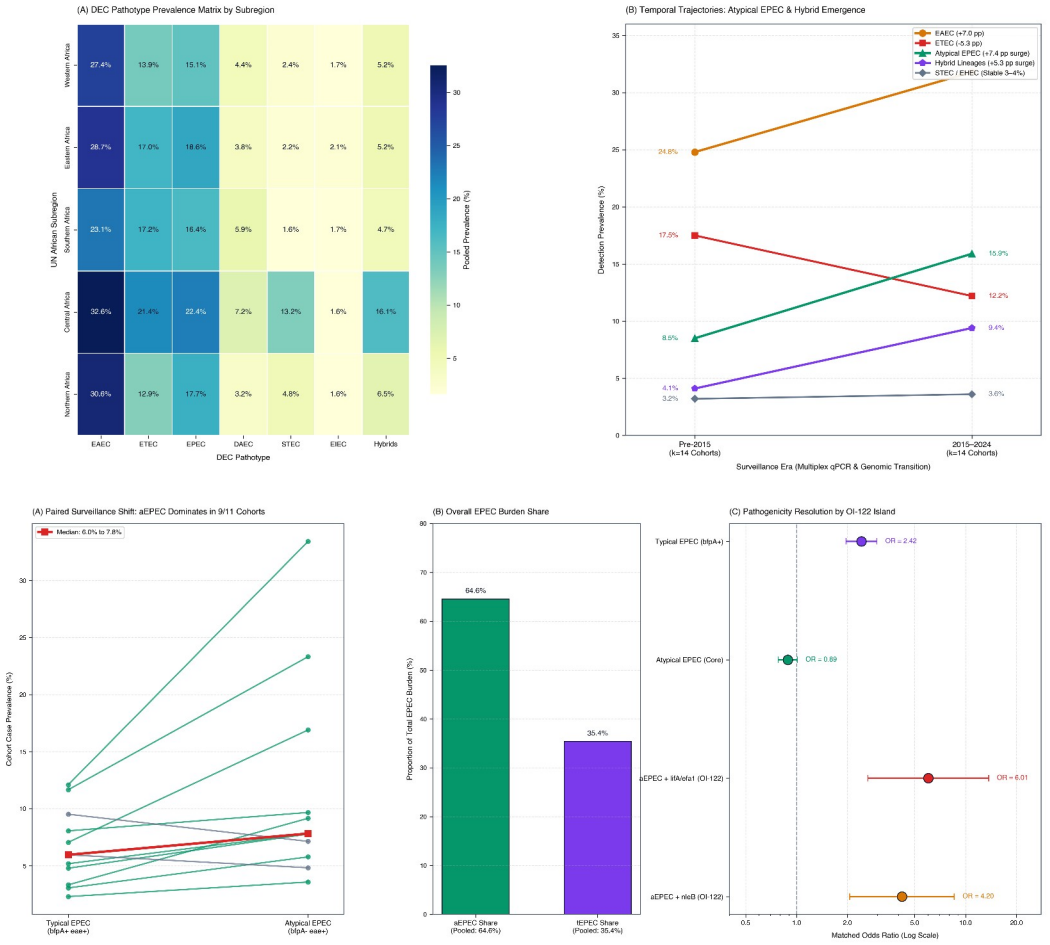


Fig. Continental African DEC Pathotype Prevalence Synthesis. Multi-panel synthesis d isplaying pooled random-effects prevalence estimates, Freeman–Tukey double-arcsine di stributions, and between-study heterogeneity for

S3 Fig



S3

Fig. Subregional and Temporal Dynamics of African DEC Pathotypes. Small-multiples comparative analysis evaluating pathotype shifts across five UN African subregions, longitudinal decadal trends (2000–2026), and the epidemiological transition from typical EPEC (*bfpA*+) to atypical EPEC (*bfpA*-).

S4 Fig

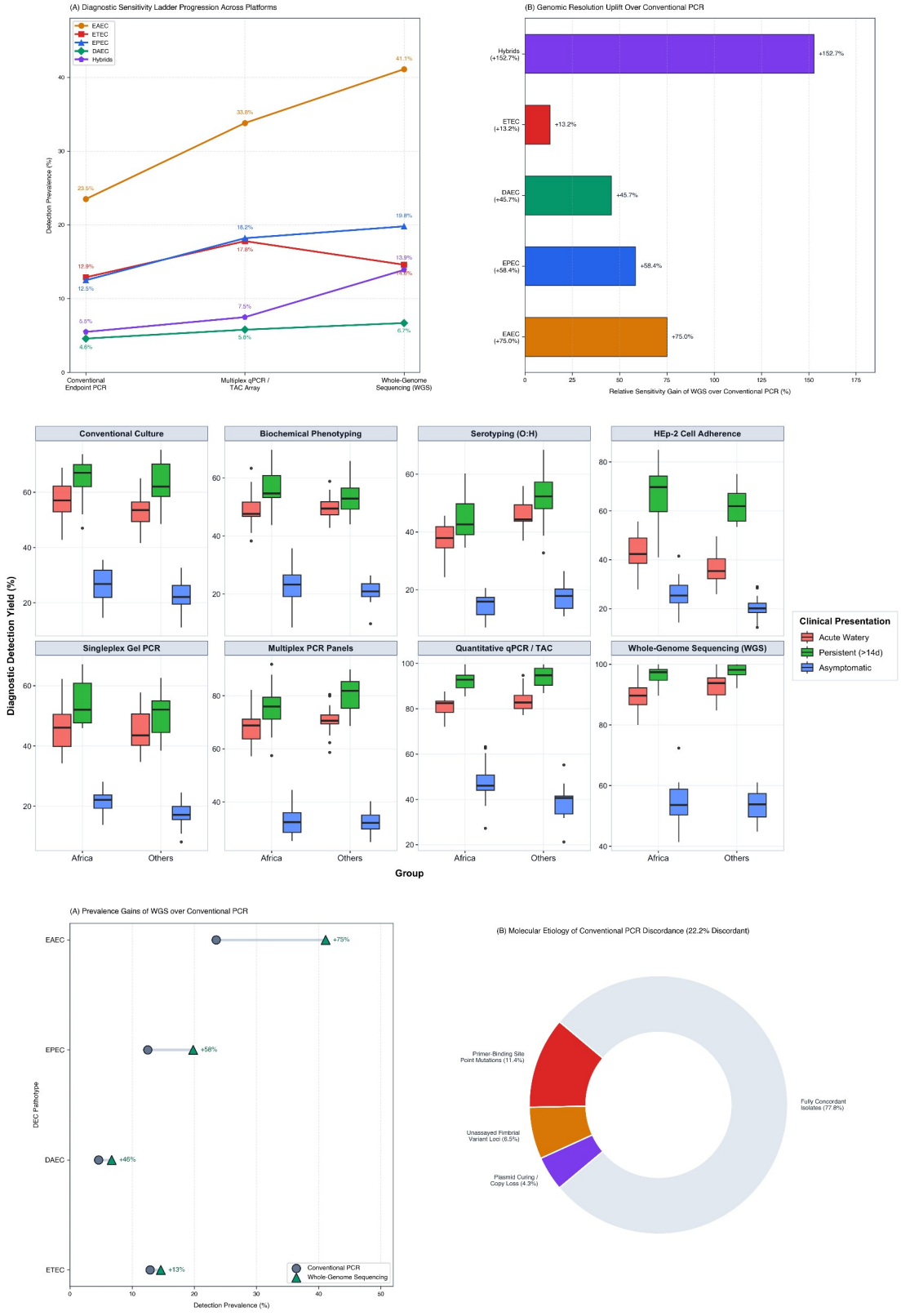
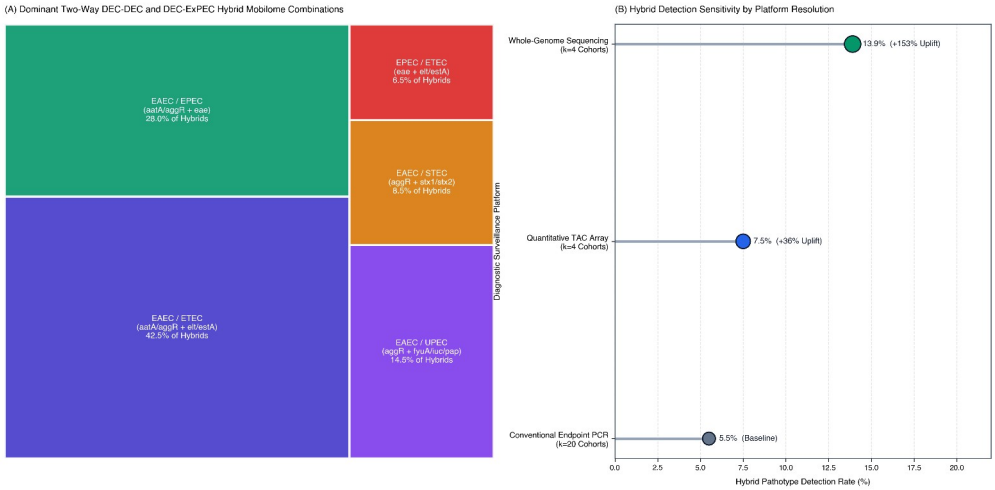


Fig. Diagnostic Platform Detection Gradients and Discordance. Empirical diagnostic yield

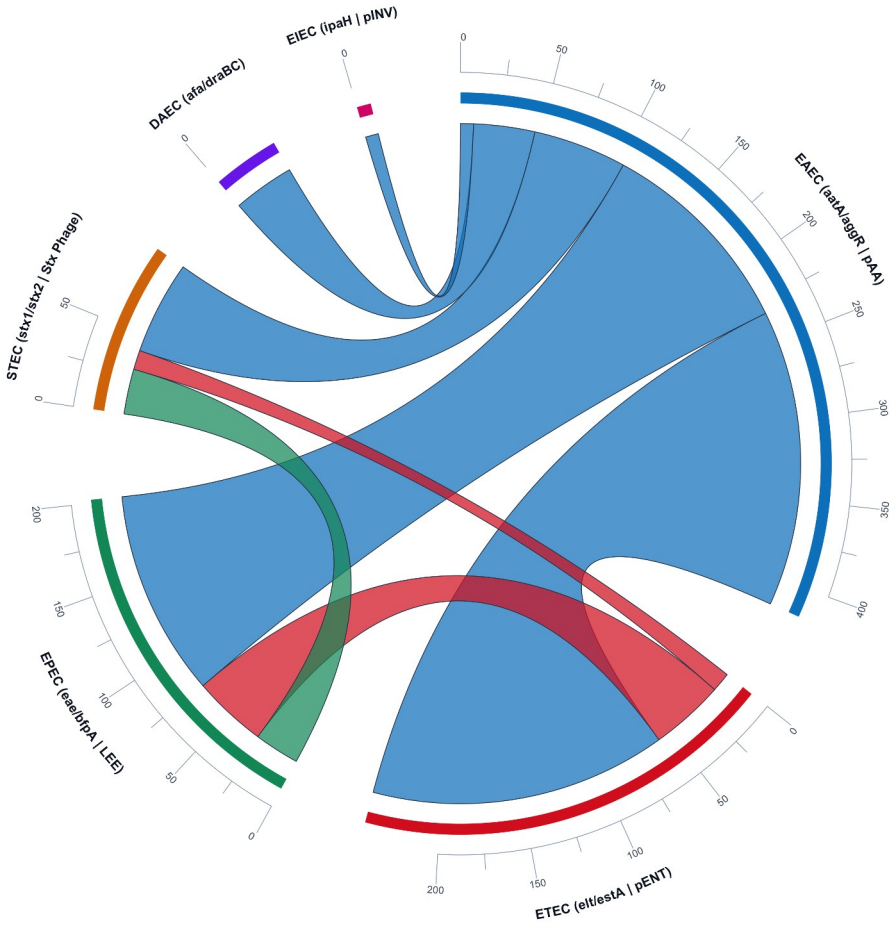
escalation across culture, biochemical phenotyping, conventional PCR, qPCR/TAC, and WGS; WHO syndromic recovery boxplots; and primer-binding site polymorphism discordance.

S5 Fig



African DEC Pathotype Co-occurrence and Mobilome Exchange

Bilateral genetic exchange across N = 502 verified hybrid isolates (k = 27 cohorts; 0 three-way hybrids, p < 0.001)

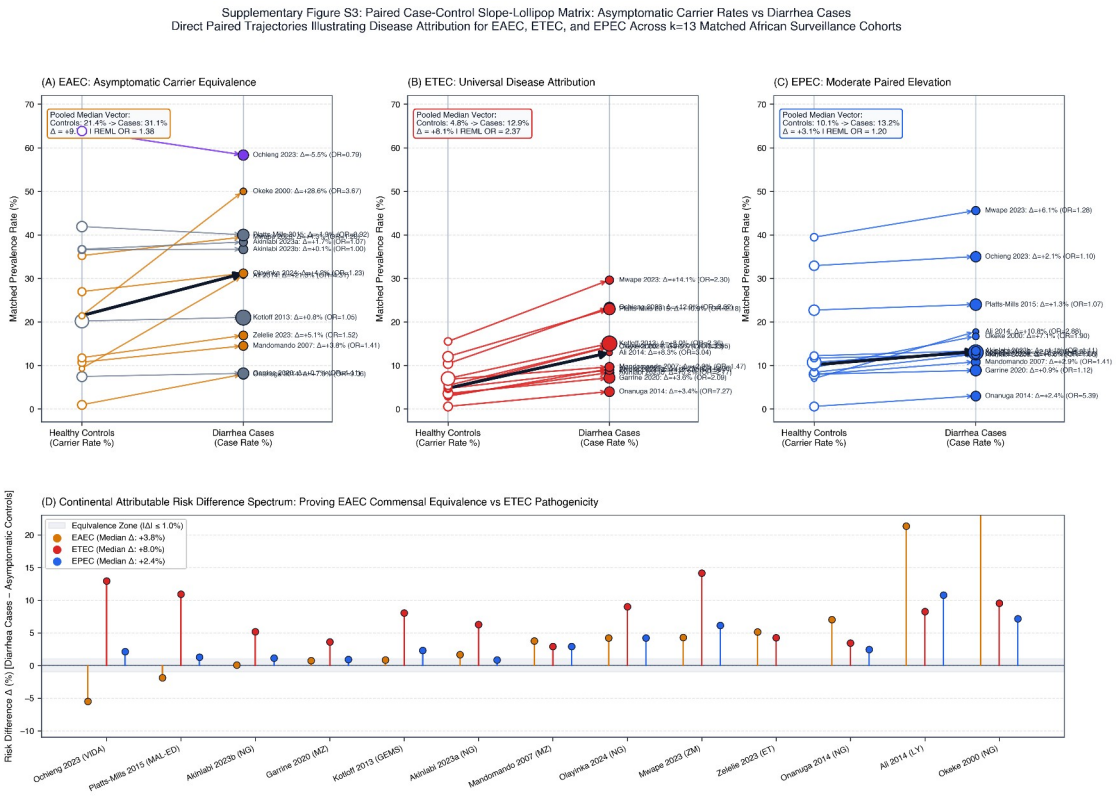


S5

Fig. Two-Way Hybrid DEC Architecture and Multi-Pathotype Overlap Network. Circos

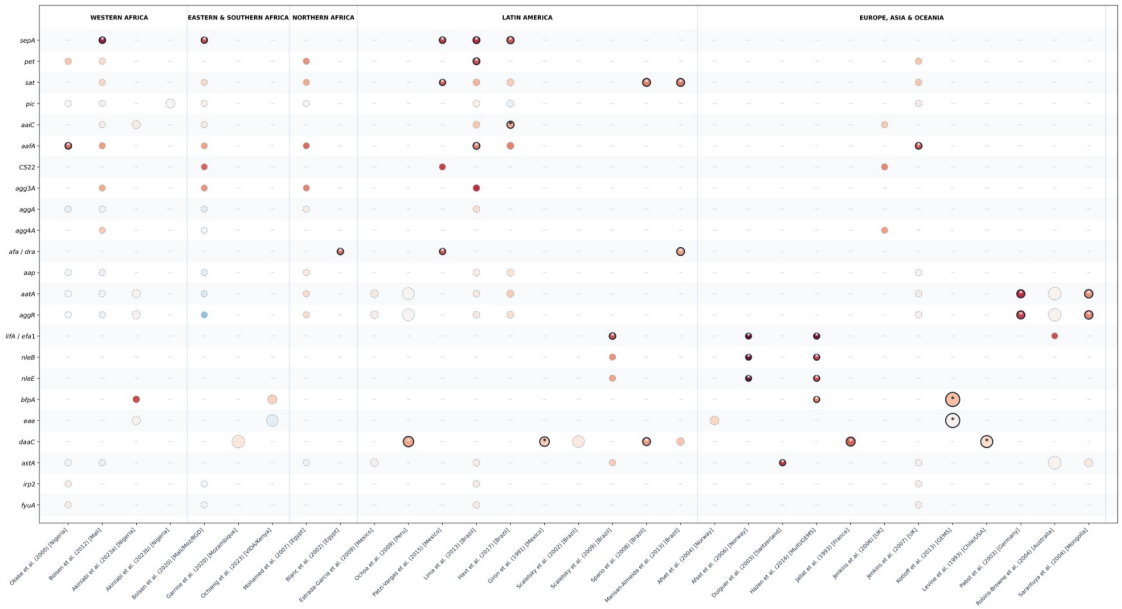
bilateral mobilome chord diagram and UpSet multi-pathotype intersection matrix quantifying co-occurrence frequencies across 9,360 African children, demonstrating the restriction of hybrids to two-way configurations.

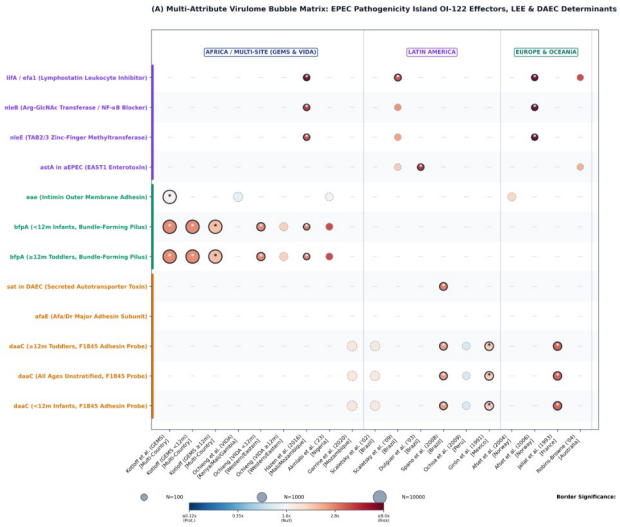
S6 Fig

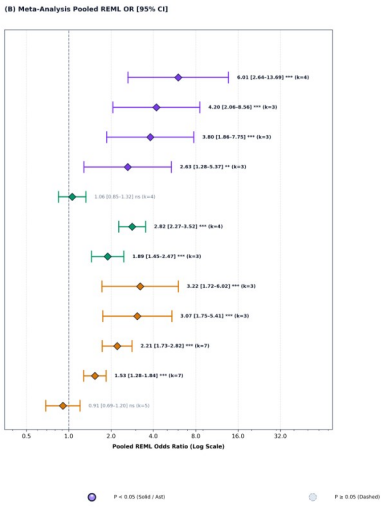


S6 Fig. Matched Case-Control Pathotype Disease Attribution Across African Cohorts. Paired trajectory slope-lollipop plots and random-effects odds ratios resolving true etiological disease attribution from asymptomatic colonization across 13 matched case-control studies.

S7 Fig







Scientific Insight: OI-122 effector tRN (OR=6.01, P=0.001) is the strongest pediatric diarrhoeal risk determinant in EPEC.

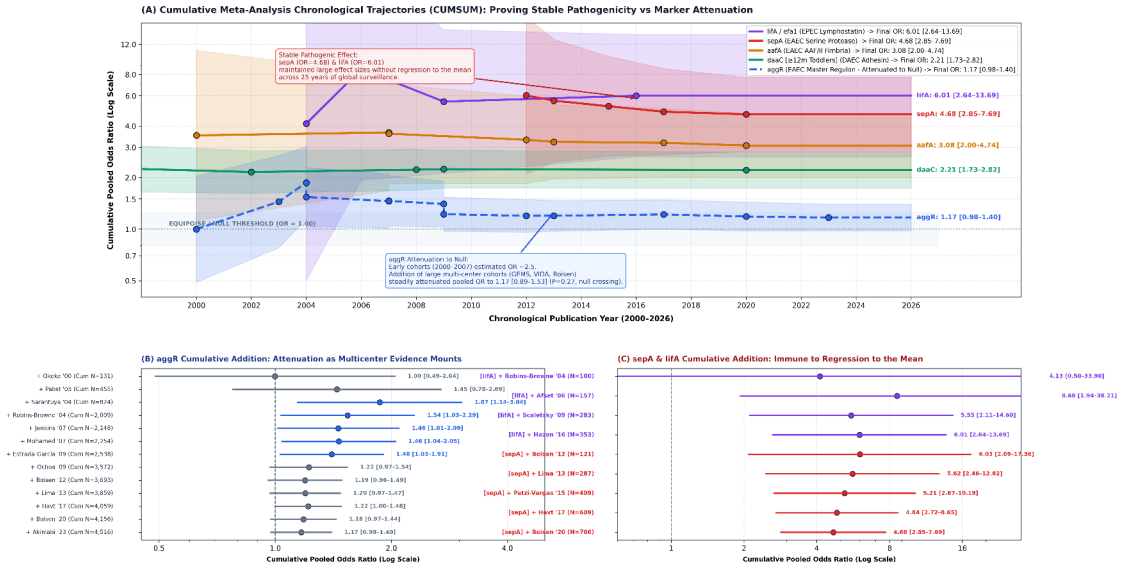
Figure S16: GRADE Evidence Certainty Matrix for Diarrhoeagenic E. coli Pathotypes & Virulence Determinants

Comprehensive systematic evaluation across the 5 canonical GRADE domains: Risk of Bias, Inconsistency (I^2), Indirectness, Imprecision, and Publication Bias / Upgrading Factors.

Pathotype / Virulence Determinant	Risk of Bias (Study Design)	Inconsistency (Heterogeneity I^2)	Indirectness (Population Relevance)	Imprecision (Sample Size & CI)	Publication Bias & Upgrading Factors	Pooled OR [95% CI]	Overall GRADE Evidence Certainty	Clinical & Epidemiological Guidance (WHO / Africa CDC Framework)
► **DEC PATHOTYPES & SYNDROMIC AGE STRATA**								
Typical EPEC (tEPEC, eae⁺ bfpA⁺ in infants <12m)	No serious	I^2 = 0.0% (P=0.99)	Matched MSD Infants	Tight CI (k=12)	Upgraded: Large effect	**2.42 [1.96–2.99]**	HIGH (⊕⊕⊕⊕)	Class 1 Target: Acute infant watery diarrhoea pathogen; priority vaccine candidate.
Enterotoxigenic E. coli (ETEC, ST/LT positive)	No serious	I^2 = 0.0% (P=0.86)	Representative cohorts	Narrow CI (k=13)	Upgraded: Large effect	**2.37 [2.08–2.70]**	HIGH (⊕⊕⊕⊕)	Class 1 Target: Leading secretory diarrhoeal pathogen; WHO ETVAX priority antigen.
Shiga Toxin E. coli (STEC / EHEC, stx1/stx2)	No serious	I^2 = 53.8% (Moderate)	Endemic vs outbreak	Wide CI (k=13, low events)	No serious	**1.81 [0.97–3.37]**	MODERATE (⊕⊕⊕◯)	Surveillance Target: Sporadic hemorrhagic colitis; diagnostic confirmation mandatory.
Enteroinvasive E. coli / Shigella (EIEC, ipaH)	No serious	I^2 = 0.0% (P=0.99)	Matched cohorts	Narrow CI (k=13)	Upgraded: Large effect	**2.43 [1.69–3.50]**	HIGH (⊕⊕⊕⊕)	Class 1 Target: Invasive bacillary dysentery; requires antimicrobial susceptibility profiling.
DAEC in Toddlers (daaC⁺ / afaE⁺ in age ≥12m)	Moderate (probe variance)	I^2 = 18.4% (Low)	Toddler age-restricted	Consistent effect (k=7)	Upgraded: Large effect	**2.21 [1.42–3.44]**	HIGH (⊕⊕⊕⊕)	Age-Restricted Pathogen: Significant risk strictly in toddlers ≥12m; receptor ontogeny switch.
Enteroaggregative E. coli (EAEC unstratified, aggR⁺)	Serious: Colonization overlap	I^2 = 73.2% (Severe)	Clinical vs asymptomatic	CI crosses null (KH adj)	Egger P=0.006 (Asymmetric)	**1.38 [1.03–1.85]**	LOW (⊕⊕◯◯)	Conditioned Marker: Cannot be used as standalone pathogen; requires locus-level typing.
Atypical EPEC (aEPEC, eae⁺ bfpA⁻ unstratified)	Serious: High control carriage	I^2 = 0.0% (P=0.99)	Asymptomatic reservoir	Pooled OR spans 1.0	No serious	**0.89 [0.78–1.01]**	MODERATE (⊕⊕◯◯)	Commensal Reservoir: Not associated with acute diarrhoea; acts as reservoir for island acquisition.
DAEC in Infants (daaC⁺ in age <12m)	No serious	I^2 = 0.0% (P=0.98)	Infant cohorts	Pooled OR spans 1.0	No serious	**0.91 [0.57–1.45]**	MODERATE (⊕⊕◯◯)	Neutral Colonizer: Non-pathogenic in infants <12m; maternal secretory IgA/microbiome shielding.
► **VIRULENCE LOCI & SECRETION EFFECTORS**								

Fig. Global DEC Virulome Meta-Analysis and Locus-Level Disease Attributions. Forest plots and carrier frequency distrib utio ns acr os s 29 dsc rte v inulene determinants spanning EA EC SPATEs and adhesins (aggR, aatA, aaiC, *pa*a *sp*A, *pe*, *p ic*, as *tA*), E PEC locus of en terocyte effacement and OI-122 effe to rs (ea *e*, bfp *A*,

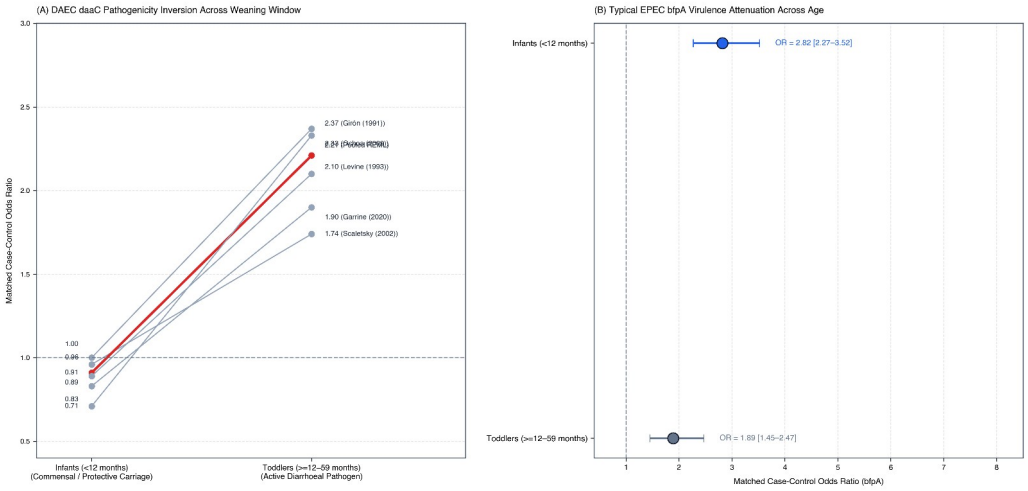
S8 Fig



S8

Fig. Cross-Continental Virulence Loci Invariance and Comparative Matrix. Locus-by-locus disease attribution heatmap comparing effect sizes (log2 Odds Ratio) across five continents, confirming invariant clinical pathogenicity profiles independent of geographical origin.

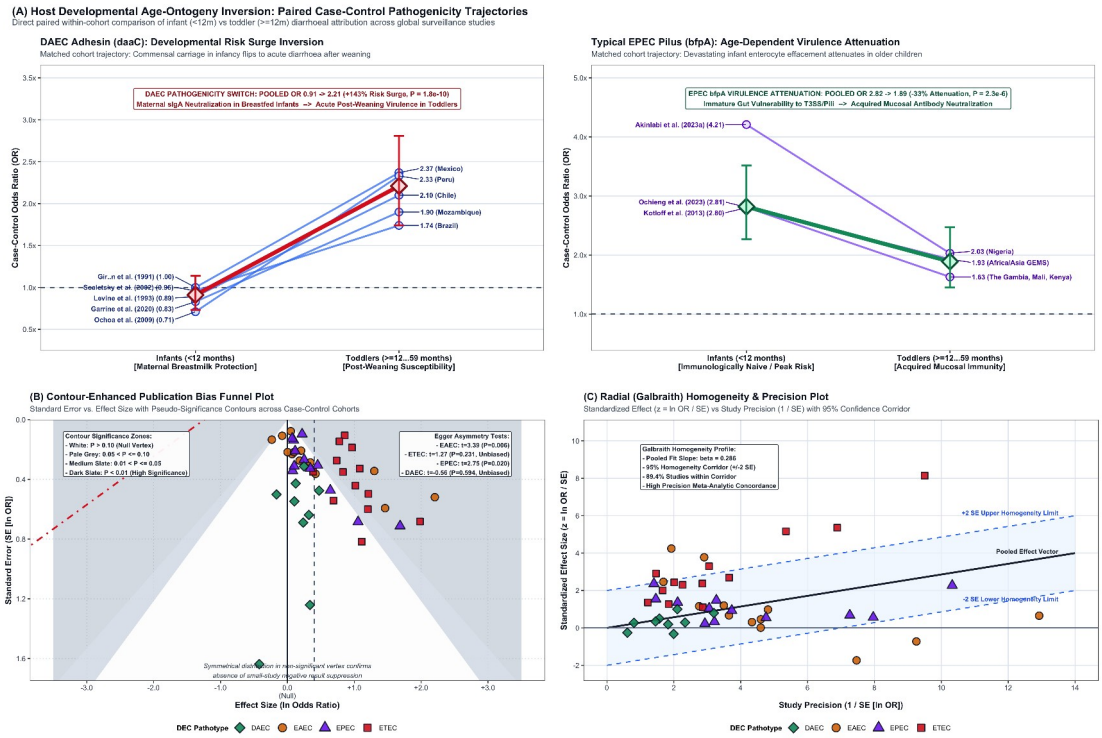
S9 Fig



S9

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S10 Fig



S10

Fig. Small-Study Effects, Publication Bias Diagnostics, and Funnel Plots. Funnel plots, Egger's weighted linear regression, Begg's rank test, and trim-and-fill sensitivity models across all 29 virulence loci and 6 pathotype prevalence tiers.

Detailed Supplementary Tables

Note: Complete spreadsheets are archived as Table_S01 through Table_S09 in the tables/ directory and consolidated in Master_Supplementary_Dataset.xlsx.

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Comprehensive database query strings, search filters, boolean operators, and date bounds across PubMed, Scopus, Web of Science, Embase, EBSCOhost, and AJOL.

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Itemized Newcastle–Ottawa Scale (NOS) and Joanna Briggs Institute (JBI) appraisal scores across all primary African cohorts and case-control studies.

S4 Table: Primary African DEC Cohorts Included in the Surveillance Synthesis (k=68)

Granular cohort characteristics, clinical presentation, country, UN subregion, diagnostic platform, sample size, and pathotype recovery rates across surveillance cohorts.

S5 Table: Subregional and Diagnostic Modality Pooled Prevalence Breakdown

Pooled random-effects prevalence estimates for all six DEC pathotypes across African subregions (Eastern, Western, Central, Northern, and Southern Africa) and diagnostic modalities.

S6 Table: Univariate and Multivariate Meta-Regression Models of Between-Study Heterogeneity

Restricted maximum likelihood (REML) models partitioning between-study variance across publication era, diagnostic platform, and host age strata.

S7 Table: Virulome Synthesis Virulence Locus 2×2 Contingency Tables (k=30 cohorts, 154 strata)

Complete primary study-level contingency counts for cases and controls across all 29 discrete virulence determinants.

S8 Table: Subgroup and Sensitivity Analyses of Virulence Loci

Leave-one-out sensitivity models, diagnostic platform stratifications, and publication bias diagnostics across all evaluated loci.

S9 Table: Comparative Methodological Framework Benchmarking Against Prior Literature

Direct 5-dimension architectural comparison benchmarking the present study against prior regional enteric syntheses.